

Harvesting risk, not return

A governed systematic platform built on an uncomfortable finding: at the horizons and instruments tested, there was no exploitable edge in *predicting* returns. The edge that survives out-of-sample lives in the *risk structure*.

by **Carter Warrens** · Blaque Baux · 2026 · reference implementation in Julia

ABSTRACT

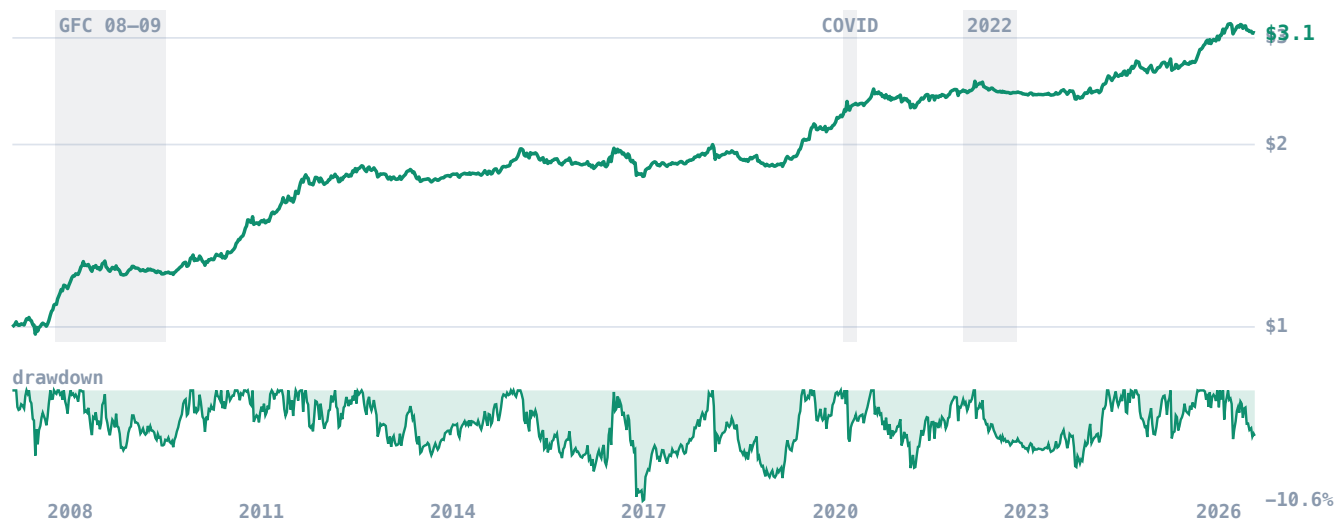
We describe a risk-premium harvesting strategy — the **spine** — that allocates by risk rather than return forecast: an inverse-volatility / equal-risk-contribution base blended with a 12-month time-series-momentum trend sleeve, each volatility-targeted on its own realized P&L, scaled by a drawdown regime brake, and rebalanced daily. On 2007–2026 (net of costs) it returns a Sharpe of ≈ 0.97 at a $\approx 11\%$ maximum drawdown, with the trend sleeve carrying the book through 2022. We motivate the design from a direct measurement — candidate return-prediction signals tested at $\approx 50\%$ accuracy (a coin flip) — and we quantify, rather than hide, the strategy's ceiling: this is single-digit-CAGR, ~ 1.0 -Sharpe money, and leverage to reach double digits buys proportional drawdown and is net-negative at today's financing rates.

1 The thesis

Most systematic strategies are sold on their ability to forecast. We tested that ability directly. On 60 days of 15-minute data across ~ 60 S&P names, two candidate alpha models — a three-factor cross-sectional model and a Bayesian Kalman/shrinkage engine — scored **50.9%** and **50.7%** directional accuracy, with information coefficients of **-0.004** and **+0.001**. Both are coin flips.

So we stopped trying to predict returns and built around what is measurable and persistent: the compensation paid for *bearing risk others avoid* — the diversification premium, and the trend (crisis-alpha) premium. Covariance is more forecastable than mean; the strategy trades on it.

FIGURE 1 – GROWTH OF \$1 AND DRAWDOWN, 2007–2026



The spine, unlevered, net of costs. \$1 → \$3.1 over ~19 years at ~6% CAGR; maximum drawdown **-10.6%** — no single crisis (GFC, COVID, 2022) worse than ~-7%. Shaded: the three crisis windows.

2 Construction

The book is two sleeves. Both are sized by an exponentially-weighted covariance estimate Σ (RiskMetrics, half-life 21), never by a return forecast.

Base sleeve — inverse-volatility weights harvest the diversification premium by putting risk, not dollars, on an equal footing (an equal-risk-contribution variant is also implemented):

$$w^{\text{base}}_i = \frac{1 / \sigma_i}{\sum_j 1 / \sigma_j}$$

Trend sleeve — a 12-month time-series-momentum sign, inverse-vol scaled into a long/short book. It goes *short* what is falling, which is why it made money in 2022:

$$s_i = \text{sign}(\Pi(1+r)_{t-251 \dots t} - 1), \quad w^{\text{trend}}_i = \frac{s_i / \sigma_i}{\sum_j |s_j / \sigma_j|}$$

Each sleeve is scaled to a target volatility using its *own realized P&L* volatility (a RiskMetrics recursion), capped — the load-bearing choice that keeps the trend hedge engaged in a crisis, when its P&L is smooth even as asset vols spike:

$$v_t^2 = \lambda v_{t-1}^2 + (1-\lambda)p_t^2, \quad e_t = \min(\text{cap}, \sigma^\star / (v_t \sqrt{\text{ppy}}))$$

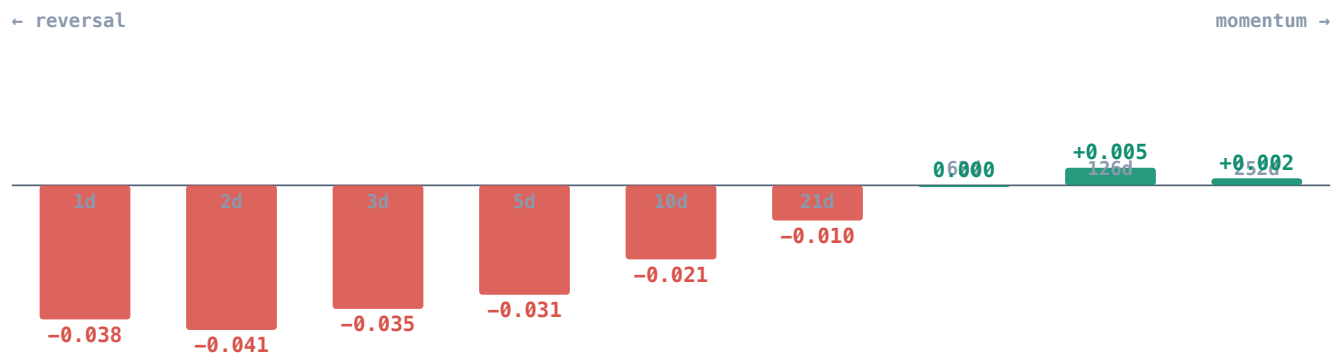
The vol-targeted sleeves are blended 50/50 and the whole book is scaled by a drawdown **regime brake** g_t that halves gross exposure when the market is >8% below its recent peak:

$$w_t = g_t \cdot (\theta e^{\text{base}} w^{\text{base}} + (1-\theta) e^{\text{trend}} w^{\text{trend}})$$

3 Why daily, and why twelve months

Rebalance cadence and lookback are not free parameters — they are dictated by where momentum actually exists. Measuring the information coefficient of a trailing move against the next day's return, by horizon, shows the market *reverses* at short horizons and only *trends* over months:

FIGURE 2 – MOMENTUM VS. REVERSAL BY HORIZON (IC, 2006–2026)

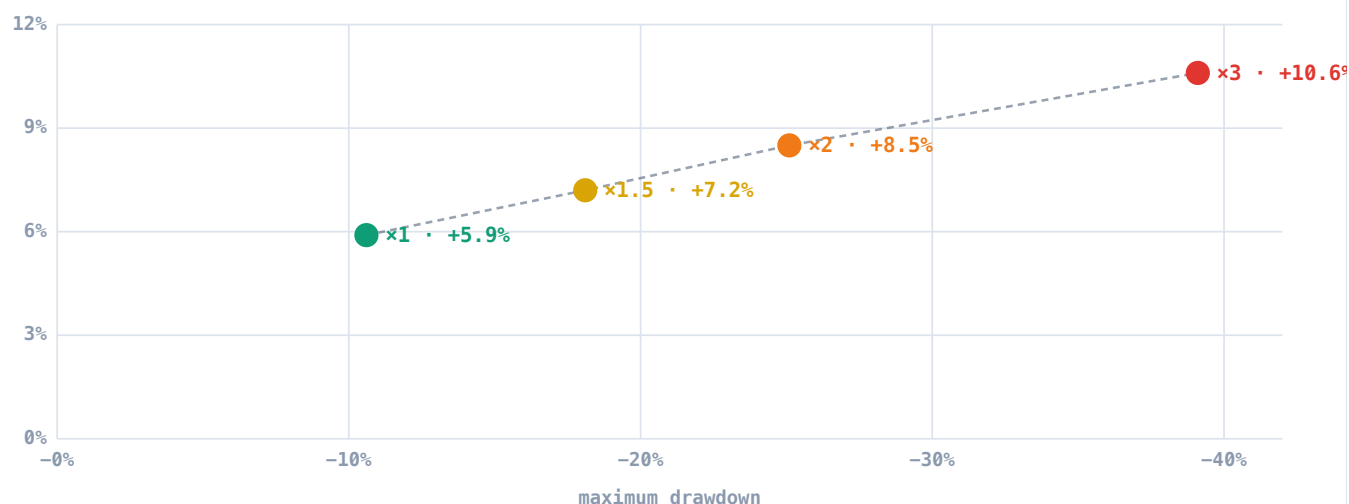


Trailing-return → next-day IC, pooled across assets. **Short horizons (1–63d) are reversal** (negative IC; 2-day is strongest); **momentum appears only at 126–252d**. This is why the trend sleeve uses a 12-month lookback and why "shorting a 2-day selloff" fights the strongest short-horizon force in the data. Reacting intraday, tested on 17,659 15-minute bars, added nothing — a diversified book absorbs intraday shocks internally.

4 The leverage trade-off, quantified

The honest ceiling. Leverage L on the strategy return, financing the borrowed portion at rate f , is $r_t^L = L r_t - (L-1)f/\text{ppy}$. Two facts follow, and neither is comfortable:

FIGURE 3 – RETURN VS. MAXIMUM DRAWDOWN BY LEVERAGE (2007–2026, 3% FINANCING)



Each point is the same strategy at a different dial setting. **Sharpe only falls with leverage** (financing is pure drag); reaching double-digit CAGR requires ~3x, which comes with a **–39% drawdown**. At today's ~6.5% margin the upper settings earn *less* than x1 — the ~6% return is below the cost of borrowing. There is no double-digit-at-low-risk point on this curve.

LEVERAGE	CAGR (3% FIN)	CAGR (6.5% FIN)	SHARPE	MAX DD
x1	+5.9%	+5.9%	0.97	-10.6%
x1.5	+7.2%	+5.4%	0.81	-18.1%
x2	+8.5%	+4.7%	0.73	-25.1%
x3	+10.6%	+3.2%	0.64	-39.1%

5 Governance, and honesty as a feature

Every order passes one governed path enforcing hard invariants — idempotency, per-pool budget and daily-loss limits, data-staleness, position reconciliation, a kill switch, and full fill lineage — beneath a pre-trade safety gate (drawdown and loss halts, gross-leverage and per-name caps). No language model is anywhere near the order path; the strategy is reproducible code.

Just as important is what we *do not* claim. The Bayesian alpha engine, the Monte-Carlo simulator, and a Sharpe-on-historical-means optimizer are all preserved but **archived**: the first measured as noise, the second is a risk calculator (it sizes a tail; it does not create return), and the third is the

Markowitz error-maximization trap. Sophisticated machinery packages and sizes a signal; it does not create one. That distinction is the whole paper.

Full mathematics, code references, and citations: [docs/FINANCIAL_METHODS.md](#). Interactive leverage explorer: [docs/leverage_decision.html](#). Architecture: [docs/CANONICAL_ARCHITECTURE.md](#). Reference implementation (Julia): [the Blaque Baux repository](#).

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